



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Mathematical Foundations of Data Science (Probability)

Course Code: CS6660

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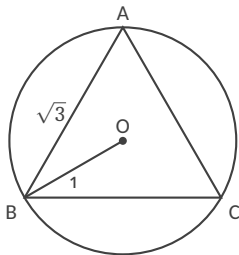
Probability Theory – Humble Beginnings

- Bernoulli (1713) and de Moivre (1718) gave the first definition of probability:

$$\text{probability of an event} = \frac{\# \text{ favourable outcomes}}{\text{total number of outcomes}}.$$

- Cournot (1843):
“An event with very small probability is morally impossible; an event with very high probability is morally certain.”
- French mathematicians of the day were satisfied with the “frequentist” approach to probability, but not the German and English mathematicians of the day
- Frequentist approach could not satisfactorily explain certain paradoxes

Bertrand's Paradox



Take a circle with unit radius and inscribe an equilateral triangle in it. Draw a “random chord”. What is the probability that the length of the chord is greater than $\sqrt{3}$?

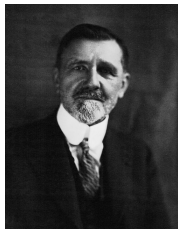
Bertrand's perfectly valid arguments:

- Mid-point of chord should lie inside incircle of radius $1/2$
Answer: $1/4$
- Tangent at A should make an angle between $\pi/3$ and $2\pi/3$ with chord
Answer: $1/3$
- Mid-point of chord should be between O and projection of O onto side BC
Answer: $1/2$

Which of the above answers is actually correct?

Borel's Measure Theory

- Contributions to Measure Theory by Borel (1894) provided satisfactory explanations to paradoxes
- A N Kolmogorov borrowed Borel's theory to formalise the axioms of probability, laying the foundation stone for modern probability theory
- For more details on the history of probability, see [[Shafer and Vovk, 2018](#)] and [[Kolmogorov, 2004](#)]



Émile Borel
(Wiki)



A N Kolmogorov and K R Parthasarathy



A N Kolmogorov

Kolmogorov's Axiomatic Theory of Probability

Sample Space, Events, Algebra, σ -Algebra, Probability Measure

Notations:

- $\mathbb{N}$: The set of all natural numbers $\{1, 2, \dots\}$
- $\mathbb{R}$: The set of all real numbers $(-\infty, +\infty)$

Sample Space

We begin with two universally accepted entities:

- Random experiment—an experiment whose result is not pre-determined
- Outcome (denoted by ω)—result of conducting the random experiment

The ONLY source of randomness in the study of probability

Definition (Sample Space)

The **sample space** of a random experiment is the set of all of its possible outcomes. We shall denote the sample space by Ω .

Example: Consider the experiment of tossing a coin once. What is Ω here?

- If our interest is in the face (head H or tail T) that lands up, then $\Omega = \{H, T\}$
- If our interest is in the velocity with which the coin lands on ground, then $\Omega = [0, +\infty) = \mathbb{R}_+$
- If our interest is in the number of times coin flips in air, then $\Omega = \mathbb{N}$

Sample Space

- Consider the experiment of repeatedly tossing a coin n times, for some fixed and finite $n \in \mathbb{N}$
What is Ω here?

- If our interest is in the sequence of face outcomes upon landing, then

$$\Omega = \underbrace{\{H, T\} \times \cdots \times \{H, T\}}_{n \text{ times}} = \{H, T\}^n$$

- If our interest is in the sequence of velocities upon landing, then

$$\Omega = \underbrace{[0, +\infty) \times \cdots \times [0, +\infty)}_{n \text{ times}} = [0, +\infty)^n$$

- If our interest is in the tuples of face outcome and the velocity upon landing, then

$$\Omega = \left(\{H, T\} \times [0, +\infty) \right)^n$$

- Consider the hypothetical experiment of repeatedly tossing a coin infinitely many times
If our interest is in the sequence of face outcomes, what is Ω ?

Answer: $\Omega = \{H, T\} \times \{H, T\} \times \cdots = \{H, T\}^{\mathbb{N}}$

Event

Informal Definition (Event)

Informally,^a an event is a subset of outcomes “of interest” to us, for which we would like to assign probabilities.

^aWe shall give a more formal definition of an event later.

- Consider the experiment of repeatedly tossing a coin 3 times
Say our interest is in the sequence of faces that show up

$$\Omega = \{H, T\}^3 = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

- A = event that at least 2 heads show up

$$A = \{HHH, THH, HTH, HHT\}.$$

Occurrence of an Event

After the random experiment is performed, an outcome ω is generated. We check what events A contain the outcome ω . We say that **A has occurred** if $\omega \in A$. Multiple events A may have occurred simultaneously for a given outcome ω .

Algebra

Definition (Algebra)

Let Ω be a sample space.

A collection $\mathcal{A}$ of subsets of Ω is called an **algebra** if it satisfies the following properties:

1. $\Omega \in \mathcal{A}$.
2. $A \in \mathcal{A} \implies A^c \in \mathcal{A}$ (closure under **complements**).
3. $A, B \in \mathcal{A} \implies A \cup B \in \mathcal{A}$ (closure under **unions**).

Property 3 above implies, by mathematical induction, that

$$A_1, A_2, \dots, A_n \in \mathcal{A} \implies \bigcup_{k=1}^n A_k \in \mathcal{A} \quad \text{for all } n \in \mathbb{N} \quad (\text{closure under all finite unions}).$$

Exercise

Show that an algebra is closed under all finite intersections.

Algebra – Examples

- Consider the experiment of throwing a six-sided dice (with faces labelled $1, \dots, 6$ once)
 - $\Omega = \{1, \dots, 6\}$
 - The smallest possible algebra on Ω is $\mathcal{A} = \{\emptyset, \Omega\}$
 - The largest possible algebra on Ω is $\mathcal{A} = 2^\Omega$ (collection of all subsets of Ω)
 - $\mathcal{A} = \left\{ \emptyset, \Omega, \{1\}, \{2, 3\}, \right.$

Algebra – Examples

- Consider the experiment of throwing a six-sided dice (with faces labelled $1, \dots, 6$ once)
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 - $\mathcal{A} = \left\{ \emptyset, \Omega, \{1\}, \{2, 3\}, \{1, 2, 3\}, \{4, 5, 6\}, \{1, 4, 5, 6\}, \{2, 3, 4, 5, 6\} \right\}$

Algebra – Insufficiency

- Consider the experiment of repeatedly tossing a coin **until the first head shows up**
 - $\Omega = \{H, TH, TTH, TTTH, \dots\}$
 - Consider the collection of all singleton subsets of Ω :

$$\mathcal{C} = \left\{ \{H\}, \{TH\}, \{TTH\}, \{TTTH\}, \dots \right\}$$

- Starting with $\mathcal{C}$, construct the collection

$$\mathcal{A} = \left\{ \emptyset, \Omega, \text{all finite unions of sets from } \mathcal{C} \right\}.$$

- Clearly, $\mathcal{A}$ is an algebra
- Suppose our event of interest is

$$A = \left\{ \text{experiment stops after even number of tosses} \right\} = \left\{ TH, TTTH, TTTTTH, \dots \right\}.$$

- Notice that $A \notin \mathcal{A}$ because A cannot be expressed as a finite union of sets from $\mathcal{C}$

σ -Algebra

Definition (σ -Algebra)

Let Ω be a sample space.

A collection $\mathcal{F}$ of subsets of Ω is called a **σ -algebra** if it satisfies the following properties:

- $\Omega \in \mathcal{F}$.
- $A \in \mathcal{F} \implies A^c \in \mathcal{F}$ (closed under **complements**).
- $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{k \in \mathbb{N}} A_k \in \mathcal{F}$ (closure under **countably infinite unions**).

Remarks:

- Henceforth, elements of a σ -algebra will be called **events**
- An event $A \in \mathcal{F}$ is also referred to as an **$\mathcal{F}$ -measurable** set
- Every σ -algebra is also an algebra, but the converse is not true
- The pair $(\Omega, \mathcal{F})$ is called a **measurable space**

Probability Measure

Fix a measurable space $(\Omega, \mathcal{F})$.

Definition (Probability Measure)

A function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is called a **probability measure** if the following properties are satisfied:

1. $\mathbb{P}(\emptyset) = 0$.
2. $\mathbb{P}(\Omega) = 1$.
3. If $A_1, A_2, \dots$ is a countably infinite collection of **mutually disjoint** sets, with $A_k \in \mathcal{F}$ for each $k \in \mathbb{N}$ and $A_k \cap A_\ell = \emptyset$ for all $k \neq \ell$, then

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) = \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**

Properties of Probability Measure

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any two **disjoint** sets $A, B \in \mathcal{F}$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any $n \in \mathbb{N}$ and a collection of **mutually disjoint** sets $B_1, \dots, B_n \in \mathcal{F}$,

$$\mathbb{P} \left(\bigsqcup_{k=1}^n B_k \right) = \sum_{k=1}^n \mathbb{P}(B_k).$$

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any set $A \in \mathcal{F}$,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **(Monotonicity:)**

For any two sets $A, B \in \mathcal{F}$ such that $A \subseteq B$,

$$\mathbb{P}(A) \leq \mathbb{P}(B).$$

Important Question

When does equality hold in the inequality above?

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any two sets $A, B \in \mathcal{F}$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **(Inclusion-Exclusion Principle:)**

For any $n \in \mathbb{N}$ and any events $A_1, \dots, A_n \in \mathcal{F}$,

$$\begin{aligned} \mathbb{P} \left(\bigcup_{i=1}^n A_i \right) &= \sum_{i=1}^n \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) + \sum_{i < j < k} \mathbb{P}(A_i \cap A_j \cap A_k) - \\ &\quad \dots + (-1)^{n+1} \mathbb{P} \left(\bigcap_{i=1}^n A_i \right). \end{aligned}$$

Continuity of Probability Measure – 1

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$, where $A_k \in \mathcal{F}$ for each $k \in \mathbb{N}$, then

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) = \mathbb{P}\left(\lim_{n \rightarrow \infty} A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

Continuity of Probability Measure – 2

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$, where $A_k \in \mathcal{F}$ for each $k \in \mathbb{N}$, then

$$\mathbb{P}\left(\bigcap_{k \in \mathbb{N}} A_k\right) = \mathbb{P}\left(\lim_{n \rightarrow \infty} A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

Union Bound

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) \leq \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

Union Bound

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) \leq \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

Corollary

For any two sets $A, B \in \mathcal{F}$ (not necessarily disjoint),

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

More generally, for any $n \in \mathbb{N}$ and sets $A_1, \dots, A_n \in \mathcal{F}$ (not necessarily mutually disjoint),

$$\mathbb{P}\left(\bigcup_{k=1}^n A_k\right) \leq \sum_{k=1}^n \mathbb{P}(A_k).$$

Probability Assignment – Examples

- $\Omega = \{H, T\}, \quad \mathcal{F} = 2^\Omega = \left\{ \emptyset, \Omega, \{H\}, \{T\} \right\}$
- $\Omega = \{1, 2, \dots, 6\}, \mathcal{F} = \left\{ \emptyset, \Omega, \{1\}, \{2, 3\}, \right.$

Probability Assignment – Examples

- $\Omega = \{H, T\}, \quad \mathcal{F} = 2^\Omega = \left\{ \emptyset, \Omega, \{H\}, \{T\} \right\}$
- $\Omega = \{1, 2, \dots, 6\}, \quad \mathcal{F} = \left\{ \emptyset, \Omega, \{1\}, \{2, 3\}, \{1, 2, 3\}, \{4, 5, 6\}, \{1, 4, 5, 6\}, \{2, 3, 4, 5, 6\} \right\}$

On Zero-Probability Sets

- $\mathbb{P}(A) = 0 \not\Rightarrow A = \emptyset$
- $\mathbb{P}(A) = 1 \not\Rightarrow A = \Omega$

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